Band-1 Hadamard Order 7732 - Miyamoto (Paley-I Seed H_1932 q=1931 + C_1934); Miyamoto-only NOT in SPK; Gap 2052 Still OPEN

H_1932 Paley-I PASS; C_1934 PASS; H_7732 Miyamoto PASS; classical SPK FAIL -> Miyamoto-only NOT in SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-16

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_7604_PaleyI`; `2026-09-16_Band1_Hadamard_7508_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_7732_miyamoto/`.
Cutoff. 2026-09-16T21:10:53Z (~5:10PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1932 and 7732 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I DIRECT seed H_{1932} = Paley-I($q = 1931$) and symmetric Paley conference C_{1934} over GF(1933) ($q = 1933 \equiv 1 \pmod 4$). Classical SPK for 7732: Paley-I **FAIL**s ($q = 7731 = 3^2 \cdot 859$ composite); Paley-II **FAIL**s ($q = 3865 = 5 \cdot 773$ NOT prime power); Kronecker **FAIL**s (4×1933). Therefore order **7732 is a Miyamoto-only bonus NOT in SPK**. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **7604 Paley-I IN SPK**. Prior SPK seeds **1876+7508 / 1872 / 1860 / 7444 / 1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / 7604 / ...** retained; Miyamoto-only bonuses **7492/7156/7108/6836/6788/6676/6484/6404/6388/6196/...** and **7732** excluded; **1932** folded into ONBOX/SPK when verified.

Abstract

Finite constructions of orders **1932** and **7732**.

- H_{1932} = Paley-I ($q = 1931$) - preferred seed (Paley-II $q = 965$ FAIL; Kronecker 4×483 FAIL)
- C_{1934} = Paley conference over GF(1933), $q = 1933$
- H_{7732} = Miyamoto(H_{1932}, C_{1934})
- Paley-I for 7732: **SKIP** ($q = 7731$ composite)
- Paley-II for 7732: **SKIP** ($q = 3865$ NOT prime power)
- Kronecker for 7732: **SKIP** ($4 \cdot 1933$ not admissible)

SPK: 7732 Miyamoto-only NOT in SPK. Seed 1932 enters SPK (Paley-I). Prior SPK **7604 / 7508 / 1876 / 1872 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_7732.py`

Results. overall_pass=true; units 127/127; wall 382.3528s; all max|HHˆT-nI|=0.

Cutoff: 2026-09-16T21:10:53Z (~5:10PM ET)

Arithmetic

- Target $n = 7732 = 4 \cdot 1933$; 1933 prime $\equiv 1 \pmod 4$ -> Miyamoto-eligible.
- Need H_{1932} and C_{1934} (Paley conference over prime field $q = 1933$).
- Seed: Paley-I $q = 1931$ (prime $\equiv 3 \pmod 4$) -> H_{1932} (Paley-II $q = 965 = 5 \cdot 193$ FAIL; Kronecker 4×483 FAIL).
- SPK: Paley-I $q = 7731$ **FAIL**; Paley-II $q = 3865$ **FAIL** (NOT prime power); Kronecker **FAIL** -> **Miyamoto-only NOT in SPK**.
- Prior: 7604 Paley-I IN SPK shipped; this ship $q = 1933 \rightarrow n = 7732$.

Constructions (verified)

Object	Method	max	HHˆT-nI
H_1932 Paley-I	Paley-I q=1931 DIRECT	0	5.3968
C_1934	Paley conf GF(1933) q=1933	0	4.9711
H_7732 Miyamoto	Miyamoto(H_1932 Paley-I, C_1934)	0	367.3892
H_7732 Paley-I	SKIP q=7731=3ˆ2*859	-	-
H_7732 Paley-II	SKIP q=3865=5*773	-	-
H_7732 Kronecker	SKIP 4*1933 not admissible	-	-

SPK closure note

Seeds include prior SPK ships through **7604** lineage (**7604** Paley-I; **1876/7508**; **1872** Paley-I; **1860/7444**; **1848/7396**; **1788**; **888/1776**; **860/1720/6884**; **1708**; **6724**; **840/1680**; **...**; **NOT** 7492/7156/7108/6836/6788/6676/6484/6404/6388/6196) + verified **1932** (Paley-I); **exclude** Miyamoto-only bonuses including **7732**. Known count 1546. Next constructive gap **2052** (OPEN).

Quantity	Value
7732 in SPK	**False** (Miyamoto-only)
1932 in SPK	**True** (Paley-I q=1931)
7604 / 7508 / 1876 in SPK	**True** (prior)
7492 / 7156 / 7108 in SPK	**False** (Miyamoto-only; excluded)

Honesty gates

- `clay_closed = false`
- GATE-04 collection **LOCKED**; no invented J
- Finite-order constructions **1932** and **7732** ONLY (verified max|HHT-nI|=0)
- **NO** full Hadamard conjecture claim; **NO** finished-ToE; **NO** RH / P!=NP / YM / Frankl / NS-solved / AC-solved
- SPK membership only via classical Paley-I / Paley-II / Kronecker (**Miyamoto alone is NOT SPK**)
- Standard constructive gap **2052** remains **OPEN**

Next optional

- **7988 Miyamoto PaleyII(q=997)** OR method-attack SPK gap **2052** OR Track B / Object 1 meter await